

W7 Question Bank

HTTP Orientation (8 Questions)

1. What is HTTP and what does it stand for? Why is it called a stateless protocol?
2. What is the main difference between HTTP and HTTPS? Why do we need HTTPS?
3. How does the request-response model work in HTTP?
4. What is the purpose of HTTP methods? Name the most commonly used ones and explain when you would use GET vs POST.
5. Why do we say POST is safer than GET for sending sensitive data? Explain with an example.
6. What are HTTP status codes? Explain the meaning of the first digit (1xx, 2xx, 3xx, 4xx, 5xx).
7. What does the status code 200, 404, 301, and 500 mean? Give real-world scenarios for each.
8. What are HTTP headers? Why do we need them? Give examples of common request and response headers.

HTML Foundation (10 Questions)

9. What is HTML and why is it called the skeleton of a webpage?
10. What is the basic structure of an HTML document? Why do we need `<!DOCTYPE html>` at the top?
11. What is the difference between `<head>` and `<body>` sections? What goes inside each?
12. Why are semantic HTML tags (like `<header>`, `<nav>`, `<article>`, `<footer>`) important? Give examples of when to use them over `<div>`.
13. What are HTML elements, tags, and attributes? Explain with an example.
14. What is the difference between block-level and inline-level elements? Give 3 examples of each and explain how they behave visually.
15. Why should we use lowercase for tag names and attribute values in modern HTML?
16. What is the purpose of alt attribute in `<img>` tag? Why is it important for accessibility and SEO?
17. What are void elements (self-closing tags)? Give 5 examples and explain why they don't need a closing tag.
18. How does nesting work in HTML? Why is proper nesting important?

HTML Forms (7 Questions)

19. What is the purpose of the `<form>` element? What are its two most important attributes and why?
20. What is the difference between the GET and POST methods in an HTML form? How does the data appear in the URL for each?
21. Why do we use the name attribute in form elements? What happens if we forget it?
22. What is the difference between `<input type="text">` and `<textarea>`? When would you use each?
23. How do `<label>` tags improve accessibility? Show the correct way to associate a label with an input.
24. What are the different types of `<input>` you know? Explain the use of `type="checkbox"`, `"radio"`, `"password"`, `"file"`, and `"submit"`.
25. Why do we use the required attribute? How does the browser behave when a required field is empty?

JavaScript Foundation (10 Questions)

1. What is JavaScript and why is it called a single-threaded language?
2. What are the different data types in JavaScript? Explain the difference between primitive and non-primitive types.
3. What is type coercion? Give an example of implicit and explicit type coercion.
4. What is the difference between var, let, and const? Why should we prefer const and let over var?
5. What are the different variable scopes in JavaScript (global, function, block)? Why does block scope matter?
6. What is hoisting? How does hoisting behave differently with var, let, and const?
7. What are functions in JavaScript? Explain the difference between function declaration and function expression.
8. What is an IIFE (Immediately Invoked Function Expression)? Why do we use it?
9. What are arrays in JavaScript? Why are they considered objects under the hood?
10. What is strict mode in JavaScript? Why should every script start with "`use strict`"?

JavaScript DOM (8 Questions)

11. What is the DOM? Why is it called a tree structure?
12. What is the difference between `document.getElementById()`, `querySelector()`, and `querySelectorAll()`?
13. How do you create a new element and add it to the DOM? Explain the steps.
14. What is the difference between `innerHTML` and `textContent`? When should you use each?
15. What is event bubbling and event capturing? Why does the order matter?
16. How do you stop event propagation? Explain `stopPropagation()` vs `stopImmediatePropagation()`.
17. What is event delegation? Why is it useful, especially with dynamically added elements?
18. How do you traverse the DOM (parent, child, sibling)? Give examples using `parentNode`, `children`, `nextElementSibling`.

JavaScript Advanced (10 Questions)

19. What is the difference between `==` and `===`? Why is `===` preferred?
20. What are arrow functions? How is 'this' behavior different in arrow functions vs regular functions?
21. What are template literals? Why are they better than string concatenation?
22. What are default parameters? How do they make functions more flexible?
23. What do the spread and rest operators do? Give one use case for each.
24. What is the 'this' keyword in JavaScript? How does its value change in different contexts (global, object, function, arrow)?
25. What is closure? Explain with a real-world example why closures are powerful.
26. What are classes in JavaScript? Are they real classes or just syntactic sugar?
27. What is the difference between `call()`, `apply()`, and `bind()`? When would you use each?
28. What are higher-order functions? Give examples like `map`, `filter`, and `reduce`.

JavaScript HTTP & Async (8 Questions)

29. What is JSON? Why is it the standard format for sending/receiving data in web apps?
30. What is the difference between `JSON.stringify()` and `JSON.parse()`?

31. What are Promises? Why were they introduced instead of using callbacks only?
32. What are the three states of a Promise? Explain with an example.
33. What is the Fetch API? How is it different from the old `XMLHttpRequest`?
34. What are `async` and `await`? Why do they make asynchronous code look synchronous?
35. How does error handling work with `async/await` vs `Promises (.catch)`?
36. What is the event loop? Why is JavaScript able to handle async operations despite being single-threaded?

JavaScript Testing (4 Questions)

37. Why do we write tests for JavaScript code?
38. What is Jest? What kind of tests can you write with it (unit, integration)?
39. What is the difference between `describe()` and `it()/test()` in Jest?
40. How do you test asynchronous code in Jest (using `async/await` or `resolves/rejects`)?

Chrome DevTools (5 Questions)

41. Why is the Console panel useful during development? Give 3 practical uses.
42. How do you find and fix performance issues using the Performance panel?
43. How can you debug network requests using the Network panel? What do status codes and timing tell you?
44. What is the difference between the Sources panel and Elements panel?
45. How do you set breakpoints in the Sources panel? Name different types of breakpoints (line, DOM, event listener).